Roll No. | | | | | | | |

END SEMESTER EXAMINATION - 2025

Name of the Course: **B.Tech**
Name of the Paper: **Automotive Mechatronics**
Time: **3 Hours**

Semester: **VI**
Paper Code: **TMS 601**
Maximum Marks: **100**

Note:-

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20marks)	
(a)	Describe the concept of high-level aggregates in automobile design, such as powertrains and suspension systems. What considerations are made during their development?	CO1
(b)	Discuss the different classification criteria for automobiles, including engine type, fuel type, size, and intended use. Provide examples for each category.	
(c)	Explain the role of chassis structures in vehicle design and performance. How do different materials influence the strength and weight of a chassis?	
Q2	(20 marks)	
(a)	Explain the importance of instrumentation gauges in a car. List and describe at least five important gauges used in vehicles and their functions.	CO2
(b)	Research and analyze the latest advancements in automobile actuators and instrumentation. How are these innovations improving vehicle performance and safety?	
(c)	Compare solenoid actuators and electric motors in terms of their working mechanism, efficiency, and typical applications in automobiles.	
Q3	(20 marks)	
(a)	Discuss the role of sensors in modern automobiles. Provide examples of how different types of sensors contribute to vehicle safety and performance.	CO3
(b)	How does an oxygen sensor function in maintaining optimal fuel-to-air ratios in the exhaust system? Discuss its impact on emission control.	
(c)	Illustrate the working of a mass air flow sensor. What challenges arise if the MAF sensor fails, and how does it affect engine performance?	
Q4	(20 marks)	
(a)	Describe the architecture and working principle of the CAN protocol. Discuss its advantages and limitations in automotive applications.	CO4
(b)	Compare CAN, LIN, Bluetooth, FlexRay, and MOST protocols. Identify scenarios in which each protocol is most suitable.	
(c)	Outline the architecture of an IoT-enabled vehicle system. Explain how various components work together to create seamless communication.	
Q5	(20 marks)	
(a)	Analyze the role of global policies and infrastructure in the adoption of fuel cell vehicles. What steps are required for widespread usage?	CO5
(b)	Evaluate the environmental impacts of hybrid, electric, and fuel cell vehicles compared to traditional internal combustion engine vehicles.	
(c)	Describe the technologies enabling autonomous vehicles, such as LiDAR, radar, and AI. How do they interact to ensure safe navigation?	